

Crossed Molecular Beam Study of $F + C_2H_4$, C_2D_4

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Crossed Molecular Beam Study of $F + C_2H_4$, C_2D_4

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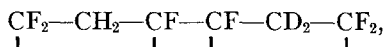
(Received 19 November 1971)

Crossed molecular beam experiments on $F + C_2H_4$, C_2D_4 have been performed with a mass spectrometric universal detector. Angular distributions of both substitution and abstraction reaction products were measured. The substitution reaction to form C_2H_3F (C_2D_3F) appears to proceed via a long-lived complex, yet the recoil energy of the products has a broad distribution peaking at about half of the energy available. Experimental results for the substitution reaction are compared with the predictions of the RRKM theory of unimolecular decomposition. The discrepancies between them and inadequacies of the theory are discussed. Highly vibrationally excited DF was produced through the abstraction reaction of $F + C_2D_4$. DF formed in the forward direction of F motion is primarily in $v' = 4$. The formation of $DF(v' = 4)$ also suggests, from the conservation of total energy, that the internal excitation of the other reaction product C_2D_3 should be very small.

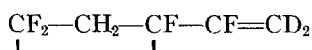
I. INTRODUCTION

A variety of approaches have recently been used to test the random lifetime assumption of the Rice-Ramsperger-Kassel-Marcus theory of unimolecular decomposition, which requires that energy be randomized among all internal degrees of freedom in the transition state. The basic questions involved are how much excess energy above the critical energy of dissociation is necessary for the assumption to break down, what types of molecular groups are more isolated from initially localized energy, and under what conditions the application of transition state theory is valid.

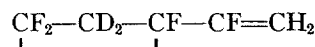
Using chemical activation techniques in the high pressure limit, Rabinovitch and co-workers have studied the constancy of the unimolecular decomposition rate for changes in the stabilization rate over several orders of magnitude for several radicals. For decomposition of butyl radicals to methyl radicals and propylene,¹ the unimolecular decomposition rate was independent of pressure for the average interval between collisions even as short as 2×10^{-13} sec, implying that the time required for energy randomization in this activated complex should be much less than 2×10^{-13} sec. However, for decomposition of



the ratio of rates for processes giving



and



depends upon pressure and also upon which ring contains the excess energy initially,² suggesting that the molecule decomposes while the energy is still localized. Monte Carlo trajectory calculations have also been used to provide information on complex lifetimes. Using a specific criterion for choosing the transition state, Bunker and Pattengill demonstrated the success of RRKM theory for idealized triatomic systems.³ On the other hand, in the more complicated rearrangement reaction of CH_3CN , RRKM theory was found to be invalid,⁴ contrary to experimental conclusions.⁵

A quite different approach has been to measure recoil energy of the products from a unimolecular decomposition where no stabilization collisions are possible. Haney and Franklin⁶ as well as Taubert⁷ have measured average recoil energy from decomposition of ions formed by electron impact. Agreement with RRKM theory is incomplete if all the internal degrees of freedom are allowed to share the energy at the transition state.⁸ For ion-molecule reactions, Wolfgang and co-workers have measured velocities and angular distributions of fragments from the long-lived complex $C_4H_8^+$, and have observed a transition to a direct reaction mechanism as the collision energy was raised.⁹ Mahan and co-workers have also studied complex formation in the ion-molecule reactions of $O_2^+ + H_2$, HD , and O_2 , and

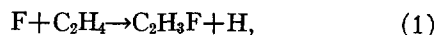
product ratios and lifetimes were compared with RRKM predictions.¹⁰ Further observations of complexes formed in ion-molecule reactions have been reviewed by Friedman and Reuben.¹¹ For the reactions of atoms and molecules, Miller, Safron, and Herschbach¹² have measured product energy and angular distributions for the crossed beam reactions $M + M'X \rightarrow MX + M'$ (M is the alkali and X is the halogen), and found substantial agreement with statistical models for decomposition of a long-lived complex. More recently, the product angular and velocity distributions were measured in many reactions involving chlorine atoms and halogenated olefin molecules, such as $Cl + C_2H_3Br \rightarrow Br + C_2H_3Cl$ by McDonald, Cheung, and Herschbach.¹³

Clearly, the applicability of RRKM theory depends upon the efficiency of intramolecular energy transfer. Despite the large number of studies concerning the energy randomization assumption, precise criteria for applicability have not been established because of the lack of knowledge of the nature of intramolecular energy transfer.

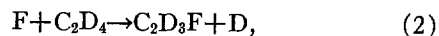
In this paper we present the results of crossed molecular beam studies of F and C_2H_4 , C_2D_4 , which are the first part of a series of systematic investigations of the theory of intramolecular energy transfer and unimolecular decomposition.

The formation of $C_2H_4F^+$ in the reaction of F with C_2H_4 is well documented. A mixture of C_2H_4 and I_2 has been used as a scavenger of thermal ^{18}F in hot atom chemistry.¹⁴ Colebourne *et al.*¹⁵ have shown that the scavenger action proceeds by the addition of F to C_2H_4 to give excited C_2H_4F , which can either decompose to $H + C_2H_3F$ or undergo collisional de-excitation and react with I_2 to give $C_2H_4FI + I$. The unimolecular decay of $C_2H_4F^+$ has also been postulated by Flores and Darwent¹⁶ to explain formation of NO and C_2H_3F from photochemical decomposition of FNO in the presence of C_2H_4 . In addition to the substitution reaction giving C_2H_3F via a long-lived complex, the other reaction channel is the abstraction reaction forming HF .

The exoergicity of the reaction,



is about 11 kcal/mole as estimated from group contributions to ΔH_f of C_2H_3F ¹⁷ and is accurate to only about ± 3 kcal/mole. ΔH_f of C_2H_4F is ~ 50 kcal lower than that of $F + C_2H_4$. The similar reaction,



is about 1.5 kcal/mole less exoergic because of zero point energy differences. For decomposition of the complex in Reaction (1), the height of the potential surface with respect to product molecules is not known, but it might differ somewhat from the activation energy of H atom addition to C_2H_4 , which is about 1.5 kcal/

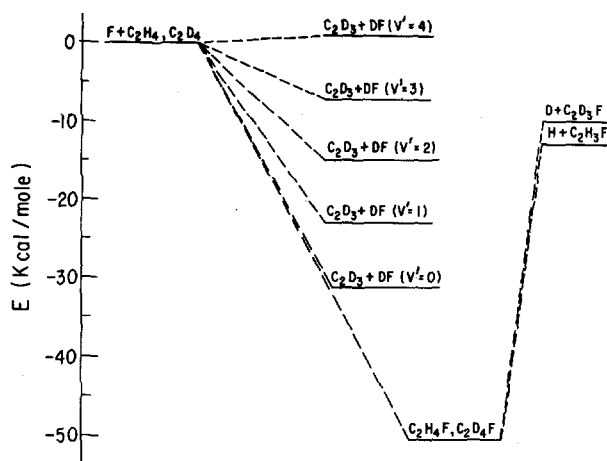
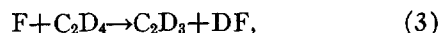


FIG. 1. Energetics for Reactions (1)–(3).

mole.¹⁸ The exoergicity of the reaction,



is approximately 31 kcal/mole, enough energy for population of the fourth vibrational level of DF when added to the typical collision energy of 2 kcal/mole. In $F + C_2D_4$ scattering, both the substitution reaction giving C_2D_3F and the abstraction reaction giving highly vibrationally excited DF were studied. The energetics for Reactions (1)–(3) are shown in Fig. 1.

II. EXPERIMENTAL

Figure 2 shows the beam arrangement of the experiment. Fluorine atoms were formed by thermal dissociation of F_2 (practical grade from Matheson Company) in a resistance heated nickel oven at around $750^\circ C$. The F_2 regulated at ~ 2 atm was first passed through an NaF trap to remove HF , and then the pressure was reduced to $\lesssim 1$ torr by a Monel needle valve. A 6 in. diffusion pump, which evacuated the oven chamber containing the fluorine oven, was separated from the chamber by a high conductance fluorine trap containing six shelves of soda lime maintained at $50^\circ C$ in order to protect the diffusion pump oil from fluorine. In the next collimation chamber, which was evacuated by a 4 in. diffusion pump, the beam was velocity selected by a slotted disk velocity selector which reduced the velocity spread to 20% full width at half-maximum. The pressures of the oven chamber and collimation chamber were $\sim 10^{-4}$ and $\sim 10^{-6}$ torr, respectively.

The ethylene beam came from a doubly differentially pumped free-jet source. Source pressures in the range 400–700 torr were used behind a 0.1 mm diam nozzle. After passing through a 0.65 mm conical skimmer, which was placed 6.0 mm from the nozzle, the beam was modulated by a 150 Hz tuning fork,¹⁹ which also

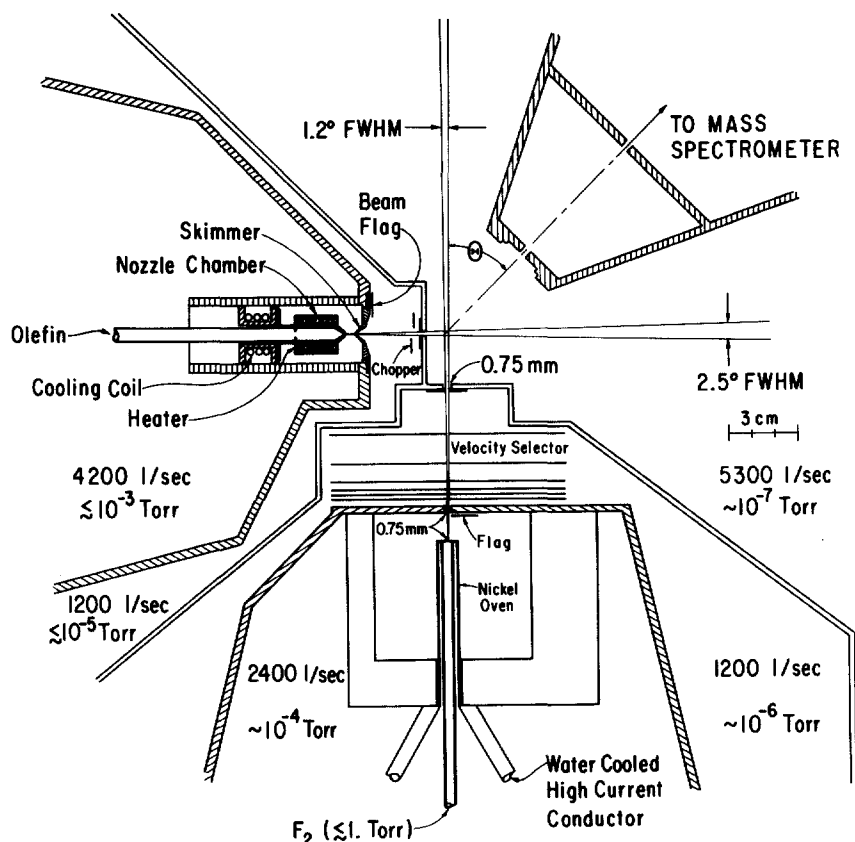


FIG. 2. Top view of beam arrangement.

supplied a reference signal to a scaler gating circuit²⁰ for background correction of the product signal. Nozzle-skimmer and skimmer-collimator chambers were evacuated by 10- and 4-in. diffusion pumps which maintained pressures in the 10^{-4} and 10^{-6} torr ranges while the beam was in operation. C_2H_4 (C.P. grade) was obtained from Matheson Company, and C_2D_4 (99% D) from Merke, Sharpe, and Dohme.

The velocity distribution of the ethylene beam was determined by temporary installation of a velocity selector and an ionization gauge beam monitor in the main chamber. The intensity of the ethylene beam corresponded to a beam pressure of $\sim 10^{-4}$ torr in the collision center. Figure 3 shows the ethylene beam flux vs velocity and selector frequency for a source pressure of 610 torr and temperature of 289°K. Calibration of the velocity selector was carried out by measuring the velocity distribution of a supersonic beam of argon, which is well known to attain a flow velocity of approximately $(5kT/m)^{1/2}$ through isentropic expansion. The theoretical velocity distribution²¹ was folded by the selector transmission function,²² and the ethylene Mach number, M , and effective specific heat ratio, γ , were varied to fit the observed distribution. Figure 3 shows the unfolded and folded calculated curves with $M = 9$, $\gamma = 1.325$ corresponding approximately to full relaxation of three rotational degrees of freedom. With the F atom beam selected at $\sim 9.1 \times 10^4$ cm/sec, the most

probable relative kinetic energy was 1.98 kcal/mole for $F + C_2H_4$ and 1.94 kcal/mole, for $F + C_2D_4$.

The beams were collimated to 1.2° and 2.5° for F and ethylene. The main chamber, where the beams crossed, was evacuated by a Norton VHS 10 diffusion pump, maintaining a total pressure of 5×10^{-8} torr. When the beams were in operation, the total pressure rose to 1.5×10^{-7} torr.

The mass spectrometric detector was similar to one described previously.²³ The main improvement involved increased pumping speed. The outermost chamber of the three nested detector chambers was evacuated by a titanium sublimation-ion pump²⁴; the middle chamber, by a 150 liter/sec differential ion pump²⁵; and the innermost chamber, which contains the ionization region, by a liquid He cryopump with a 50 liter/sec differential ion pump.²⁵ Pressures during operation were $\sim 1 \times 10^{-9}$, $\sim 5 \times 10^{-9}$, and $\leq 10^{-10}$ torr for the three chambers, respectively.

For Reaction (1), mass 46 (C_2H_3F) was counted by a gated dual counter which gives the signal plus background in one channel and the background in the other channel, for 80 sec periods with the F beam flag open and closed. One angle was used as a time normalization reference; the entire angular range was scanned repeatedly in order to minimize statistical fluctuations in the signal. Optimum conditions obtained were about 60 counts of signal per second at the most favorable

angle, and no signal correction was found necessary with the F beam flag closed except within about 20° of the ethylene beam. Later, it was found that the signal with the F beam flag closed could be further reduced by collimating the ethylene beam before the chopper. For the estimation of the total reactive cross section, elastic scattering of F atoms was measured within 10° of the F beam. In order to minimize the contribution at mass 19 from F_2 which fragments in the mass spectrometer, the fluorine source pressure was lowered in the elastic scattering measurements to optimize dissociation of F_2 in the oven. For the reaction $F + C_2D_4$, mass 49 (C_2D_3F) was counted for 80 sec periods, also, but with fewer repeated scans. In addition, a much weaker mass-21 signal (DF) was measured for 200 sec periods with repeated scans.

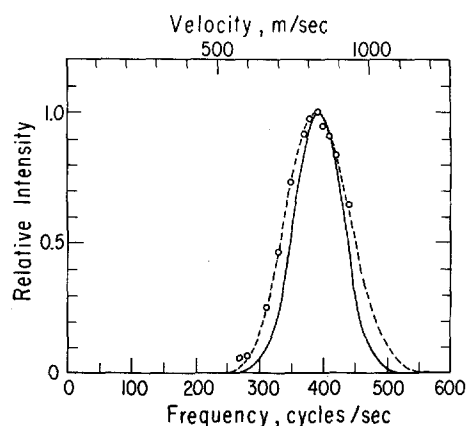


FIG. 3. $\circ$, Measured velocity distribution of ethylene beam; —, theoretical distribution with $M=9.0$, $\gamma=1.325$; ---, theoretical distribution folded with velocity selector transmission function.

III. RESULTS AND KINEMATIC ANALYSIS

The angular distributions measured for formation of C_2H_3F and C_2D_3F are shown in Figs. 4 and 5 along with nominal Newton diagrams. For C_2H_3F , error bars were estimated by computing the standard deviation of the signal at each angle in repeated scans. Both systems are seen to show forward-backward symmetry with respect to the center of mass, consistent with the formation of a long-lived complex.

For reaction products from a long-lived complex, the center-of-mass (c.m.) velocity distribution is independent of c.m. scattering angle, and the c.m. angular distribution depends on the partitioning of angular momentum. In order to obtain a c.m. velocity distribution from our angular scan, it was assumed that the c.m. angular distribution is isotropic for these systems. This assumption can be justified by consideration of the angular momenta involved. If the final orbital angular momentum L' of product molecules is a small fraction of the total angular momentum J , then v' , the relative velocity vector of reaction products from a long-lived complex, is likely to lose any correlation with v , the

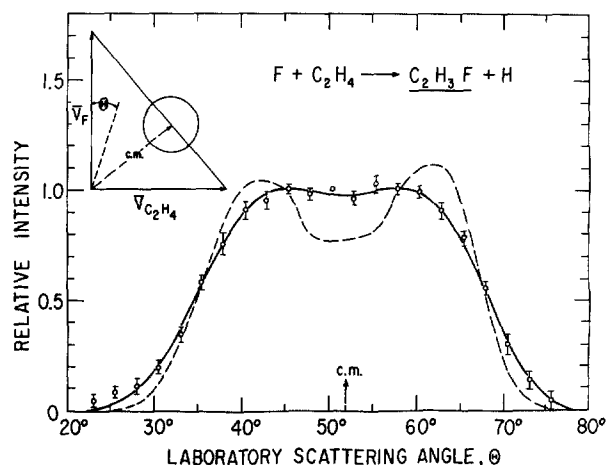


FIG. 4. $\circ$, Experimental laboratory angular distribution for Reaction (1); —, best fit laboratory angular distribution using Eq. (4) for the c.m. recoil-energy distribution; ---, laboratory angular distribution from RRKM recoil-energy distribution.

relative velocity vector of reactants. This has been demonstrated rigorously for the specific case of a symmetric top complex with statistical equilibration of rotational energy, by Miller, Saffron, and Herschbach,¹² who showed that the angular distribution becomes

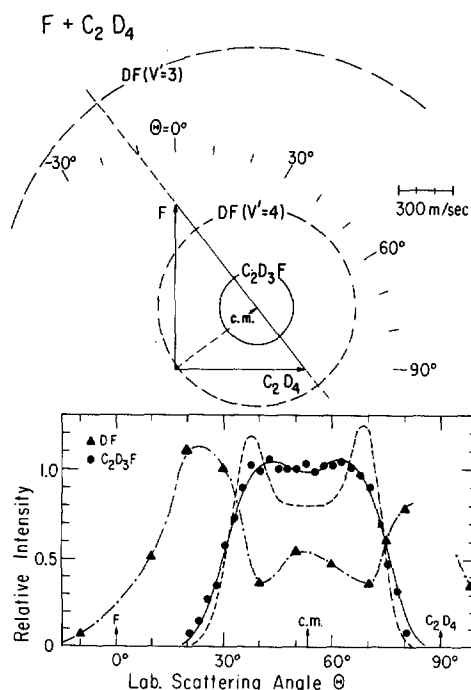


FIG. 5. Newton diagram and laboratory angular distributions for $F + C_2D_4$ reactions. The Newton diagram shows the c.m. velocity limits for DF formed in $v'=3, 4$ and the average c.m. velocity observed for C_2D_3F . Calculated laboratory angular distributions for C_2D_3F are compared with experiment. —, Best fit angular distribution using Eq. (4); ---, angular distribution from RRKM recoil-energy distribution.

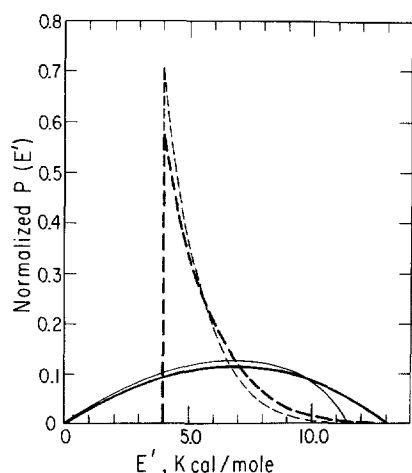


FIG. 6. The c.m. recoil-energy distributions for Reactions (1) and (2). Heavy solid line, best fit distribution for (1); heavy dashed lines, RRKM distribution for (1); —, best fit distribution for (2); ---, RRKM distribution for (2).

isotropic if the ratio of initial orbital angular momentum to the projection of total angular momentum along ∇' is small.

In the decomposition of $\text{C}_2\text{H}_4\text{F}(\text{C}_2\text{D}_4\text{F})$ into $\text{C}_2\text{H}_3\text{F}(\text{C}_2\text{D}_3\text{H})$ and $\text{H}(\text{D})$, the final orbital angular momentum $\mathbf{L}'$ can be considered either to arise from rotational angular momentum of the complex or to be created by repulsive forces which occur after the transition state for separation. The fraction of total angular momentum $\mathbf{g}$, which can be carried away by $\mathbf{L}'$, is relatively small in this case due to the small reduced mass of product molecules and the short range interaction between them. Even for the most favorable situation, when $\mathbf{g}$ is directed along the axis with the smallest moment of inertia ($\geq 12 \text{ amu} \cdot \text{\AA}^2$ for $\text{C}_2\text{H}_4\text{F}$ and $\geq 18 \text{ amu} \cdot \text{\AA}^2$ for $\text{C}_2\text{D}_4\text{F}$), the maximum $\mathbf{L}'$ for a reasonable C-H separation of $\sim 2 \text{ \AA}$ in the transition state, will be only around $\frac{1}{2}\mathbf{g}$ for H emission and $\frac{1}{3}\mathbf{g}$ for D emission. If $\mathbf{g}$ is oriented along the other axes of the complex (moments of inertia $\geq 50 \text{ amu} \cdot \text{\AA}^2$) the fraction of $\mathbf{g}$ going to $\mathbf{L}'$ is considerably smaller. This is in sharp contrast to three-atom systems such as alkali atom and alkali halide where the long range interaction between product molecules and comparable atomic masses make it possible to transfer almost all of $\mathbf{g}$ into $\mathbf{L}'$. Repulsive forces after the transition state can create $\mathbf{L}'$ and final molecular rotational angular momentum $\mathbf{J}$ when the line of forces acting between separating atom and molecule makes a finite distance from the center of mass of the molecule. Energetically, this corresponds to partial conversion of potential energy into translational and rotational energies. In polyatomic molecules, such as the systems under consideration, the spatial distribution of $\mathbf{L}'$ created by repulsive forces will be related only to the molecular orientation of the bond to be broken and not to $\mathbf{L}$ or $\mathbf{g}$, no matter how large $\mathbf{L}'$ is

with respect to $\mathbf{L}$ or $\mathbf{g}$. Although certain orientations of $\mathbf{L}$ with respect to molecular orientation may be favored for formation of the complex, the C-H bond to be broken is still most likely isotropically distributed with respect to $\mathbf{L}$ because of the presence of some initial rotational angular momentum of ethylene. Thus, even if repulsive forces dominate, the products will also be isotropically distributed. A final check of the isotropic assumption could be obtained from velocity analysis of the products.

For analysis of the product c.m. translational energy, a flexible functional form for the c.m. energy distribution $P(E')$ was used in order to fit the observed angular distributions,

$$P(E') \sim (E' - V_a)^{(n-1)/2} (E_{\text{tot}} - E')^{s-1}, \quad V_a < E' < E_{\text{tot}}, \quad (4)$$

with

$$P(E') = 0 \text{ for } E' \leq V_a \text{ and } E' \geq E_{\text{tot}}.$$

E_{tot} is the total energy available with respect to the zero point energy level of the products, i.e., initial c.m. energy of reactants plus exothermicity; V_a , n , and s are parameters, whose physical significance will be discussed later. The c.m. cross sections were transformed to the lab and averaged over beam velocity spreads for comparison with experiment. The averaging was performed by constructing 30 weighted Newton diagrams from six equally spaced F atom velocities and five equally spaced ethylene velocities, transforming the cross section for each to specified lab angles and velocities of the product, and then averaging over product lab velocities for fixed lab angle. Normalization was adjusted by a least squares method.²⁶

Three values for n were tested: 1, 2, and 3; the best fit obtained for both $\text{C}_2\text{H}_4\text{F}^\dagger$ and $\text{C}_2\text{D}_4\text{F}^\dagger$ was with $n=3$. For $\text{C}_2\text{H}_4\text{F}^\dagger$ the optimum parameters were $s=1.9$ and $V_a=0.0$; for $\text{C}_2\text{D}_4\text{F}^\dagger$, $s=1.7$, and $V_a=0.0$.²⁷ Laboratory angular distributions calculated from these recoil-energy distributions are shown in Figs. 4 and 5; the energy distributions are shown in Fig. 6. Figure 7(a) illustrates the sensitivity of the angular distribution of $\text{C}_2\text{H}_3\text{F}$ to changes in s for fixed V_a and n . Another parametric form for $P(E')$ was also tested, but proved to be less satisfactory,

$$P(E') \sim (E' - V_a)^{(n-1)/2} \exp(-E'/E_0), \quad V_a < E' < E_{\text{tot}}, \quad (5)$$

with

$$P(E') = 0 \text{ for } E' \leq V_a \text{ and } E' \geq E_{\text{tot}}.$$

For E_0 in a range necessary to fit the experimental results, the shape of the function becomes more sensitive to the cutoff energy, E_{tot} , than to E_0 .

The total cross section for Reaction (1) was determined by normalizing the $\text{C}_2\text{H}_3\text{F}$ signal to small angle F elastic scattering signal, which followed the usual $\theta^{-7/8}$ c.m. angular dependence for $1/r^6$ dispersion poten-

tials.²⁸ The elastic scattering cross section was put into absolute units by estimating the C_6 constant for $F-C_2H_4$ by the Slater-Kirkwood approximation²⁹ using polarizabilities of F ($0.6 \text{ \AA}^3$)³⁰ and C_2H_4 ($4.26 \text{ \AA}^3$).³¹ The value obtained for C_6 was $7.25 \times 10^{-59} \text{ erg} \cdot \text{cm}^6$. Integration of the best fit c.m. differential cross section yielded a reactive cross section of $2.2 \text{ \AA}^2$, after corrections were made for the relative ionization efficiencies of F and C_2H_3F (assumed proportional to polarizability), and for fragmentation of C_2H_3F in the mass spectrometer (26% of ionized C_2H_3F appearing at $m/e=46$).³² The estimated accuracy is only $\pm 50\%$.

Figure 5 shows the DF laboratory angular distribution from Reaction (3) together with the Newton diagram showing the c.m. velocity limits of DF formed in different vibrational states. Distinct forward and backward peaks of DF were observed in the laboratory angular range which corresponds to a very small product translational energy, $\lesssim 1 \text{ kcal/mole}$. Scattering in the forward direction with respect to the c.m. F velocity is favored over backward scattering by $\sim 3:2$. The location and the sharpness of the forward peak suggests that DF molecules formed in the forward hemisphere are exclusively in $v'=4$. If a significant fraction of DF ($v'=3$) were present, the peak should extend out much further from the c.m., since energetically, DF ($v'=3$) could appear as far as around -17° in lab scattering angle, and it is not likely that the sum of rotational excitation of DF ($v'=3$) and internal excitation of C_2D_3 will be sharply peaked at $\sim 8 \text{ kcal/mole}$, which is one vibrational quantum of DF. The falloff observed in the angular distribution is in good agreement with the DF ($v'=4$) threshold angle of 9° . The result that only the highest possible vibrational state of DF, $v'=4$, is formed in the forward direction is also deduced in the crossed molecular beam study of



although lower vibrational states of DF were observed in the backward direction.³³ The backward-scattered DF molecules observed from Reaction (3) are also expected to be in $v'=4$, following the same reasons given for the forward peak. If lower vibrational states of DF were also formed in Reaction (3), the only possible recoil direction would be into the backward hemisphere where our limited angular range allows us to detect only DF ($v'=4$). By analogy with the results for Reaction (6), we expect the backward-scattered lower vibrational states of DF from Reaction (3) to be comparable to or less than backward-scattered DF ($v'=4$). Thus, a substantial fraction of the over-all product from Reaction (3) is probably in $v'=4$, which limits the internal excitation of C_2D_3 to only 1–2 kcal/mole by conservation of energy. This extreme asymmetry of internal excitation of product molecules observed, the highly inverted vibrational population of DF, at least in the forward direction, and the lack of

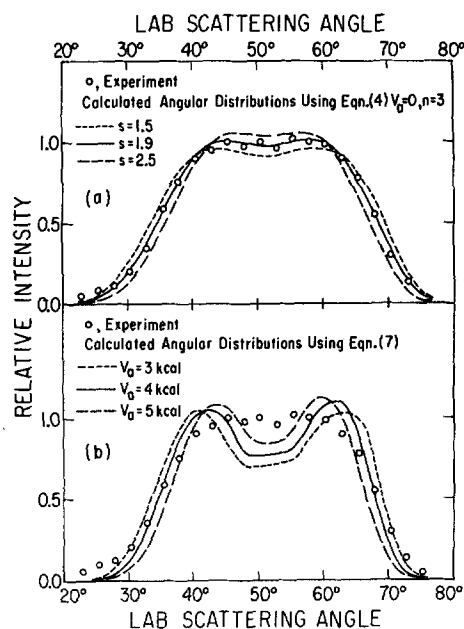


FIG. 7. (a) Comparison of experimental C_2H_3F lab angular distribution with angular distributions calculated using Eq. (4) for recoil-energy distributions with V_a and n fixed and s varied about its optimum value. (b) C_2H_3F lab angular distributions calculated using RRKM recoil-energy distribution with V_a varied.

forward-backward symmetry provide a consistent picture of Reaction (3) proceeding without formation of a long-lived complex.

IV. RRKM CALCULATION OF DECOMPOSITION OF LONG-LIVED COMPLEXES, $C_2H_4F^+$ AND $C_2D_4F^+$

The RRKM theory for unimolecular decomposition³⁴ can predict the probability of energy ϵ in the reaction coordinate at the transition state if it is possible to enumerate the density of states available to the reaction coordinate and the density of states of all other active degrees of freedom at energy $\epsilon^\dagger - \epsilon$, where $\epsilon^\dagger$ is the total internal energy available. In order to obtain the product recoil-energy distribution from the reaction coordinate energy distribution at the transition state, further assumptions are necessary: (1) No mixing of energy between the reaction coordinate and other degrees of freedom is allowed after passing the transition state; (2) effective potential energy curves can be constructed for the reaction coordinate with an accurate knowledge of the potential energy difference between the transition state and product molecules.

The density of states of the reaction coordinate at any spatial point can be easily defined if motion along the reaction coordinate is classical. In this case, $N_{rc}(\epsilon) \sim \epsilon^{1/n-1}$ for an n -dimensional reaction coordinate. For quantum mechanical motion along the reaction coordinate, an average density of states in a neighborhood

δr of the transition state can only be defined for $\delta r > \lambda$, the deBroglie wavelength for reaction coordinate motion,³⁶ but this density will depend upon the detailed shape of the potential surface throughout δr . For H emission from $\text{C}_2\text{H}_4\text{F}^\ddagger$ the potential surface cannot be easily approximated over the necessary range, since estimates of the average ϵ indicate that $\lambda \sim 1$ Å. Thus, the usual classical density of translational states was used with $n=1$ in the following calculation.

The density of states of vibrational degrees of freedom at the transition state, $N_v(\epsilon^\ddagger - \epsilon)$, was enumerated by the approximation of Whitten and Rabinovitch.³⁶ Frequency assignments for the transition states were taken primarily from frequencies for $\text{C}_2\text{H}_3\text{F}$ and $\text{C}_2\text{D}_3\text{F}$.³⁷ C-H (C-D) bending frequencies in the transition state were assumed to be 300 cm^{-1} (210 cm^{-1}),³⁸ the C=C stretching frequencies were lowered by 6%; no active rotations were assumed. The resulting zero point energies were 27.54 kcal/mole for $\text{C}_2\text{H}_4\text{F}^\ddagger$ and 21.50 kcal/mole for $\text{C}_2\text{D}_4\text{F}^\ddagger$. The β parameters were 1.3967 for $\text{C}_2\text{H}_4\text{F}^\ddagger$ and 1.3377 for $\text{C}_2\text{D}_4\text{F}^\ddagger$.

The height of the potential energy of the transition state relative to products, V_a , had to be taken as an adjustable parameter because of lack of precise knowledge of this quantity. For H(D) emission, since the departing atom recoils from a heavier atom which is located very close to the c.m., most of V_a is converted to translational energy of the products, making the final kinetic energy of the products $E' = \epsilon + V_a$. The contribution of orbital angular momentum of the products at the transition state to the effective potential energy which goes into product recoil energy was estimated to be less than ~ 0.5 kcal/mole for both H and D emission, following the argument in the previous section that only a small fraction of the total angular momentum, $J \approx L_{\text{max}} = \mu v (Q_r/\pi)^{1/2} = 18\hbar$,³⁹ can enter as product orbital angular momentum L' .

The resultant recoil energy distribution $P(\epsilon + V_a)$ is proportional to $fN_{\text{rc}}(\epsilon)N_v(\epsilon^\ddagger - \epsilon)$, where $f \propto \epsilon^{1/2}$ is the frequency of passing through the transition state. Since $E' = \epsilon + V_a$ and $E_{\text{tot}} = \epsilon^\ddagger + V_a$,

$$P(E') \sim N_v(E_{\text{tot}} - E'), \quad V_a < E' < E_{\text{tot}}, \quad (7)$$

with

$$P(E') = 0, \quad E' \leq V_a \text{ and } E' \geq E_{\text{tot}}.$$

The laboratory angular distributions were calculated from this form of the recoil-energy distribution assuming an isotropic c.m. angular distribution as described in the previous section. It was found that $V_a = 4$ kcal/mole was necessary for both $\text{C}_2\text{H}_4\text{F}^\ddagger$ and $\text{C}_2\text{D}_4\text{F}^\ddagger$ to make the laboratory angular distributions wide enough to match the edges of the experimental angular distributions. The laboratory angular distributions are shown in Figs. 4 and 5, and the RRKM c.m. energy distributions used are compared in Fig. 6 with the best fit energy distributions described in the previous section. Figure

7(b) shows how the calculated angular distribution of $\text{C}_2\text{H}_3\text{F}$ changes with V_a .

V. DISCUSSION

In contrast to pressure dependence studies of intramolecular energy transfer and unimolecular decomposition, the analysis of molecular beam results is not complicated by collisional stabilization efficiencies. In fact, essentially only one channel is open to the long-lived fluoroethyl complex once it is formed. Decomposition back to reactants is negligible because the collision energy is small compared with the exothermicity of the substitution reaction. Consequently, the cross section of the substitution reaction, 2.2 Å^2 , corresponds to the cross section for formation of the collision complex. Since there is probably no activation energy for formation of the collision complex, such a small cross section might imply a preferred orientation of C_2H_4 with respect to the approach of the F atom for formation of the collision complex. For the abstraction reaction to form DF, the analysis of the laboratory angular distribution clearly indicates that the abstraction of D (or H) does not proceed via the same long-lived complex. Thus, Reactions (2) and (3) compete only during the approach of the F atom to C_2D_4 . Otherwise, they are uncoupled. It is likely that DF formation is preferred if the F atom approaches C_2D_4 along the direction of the C-D bond, while formation of the collision complex leading to the substitution reaction may result if the F atom approaches the delocalized π electrons in the direction perpendicular to the molecular plane of C_2D_4 . Reactions (2) and (3) will be discussed separately.

A. Substitution Reaction

For the substitution reaction (1) or (2) the angular distribution obtained from the RRKM recoil-energy distribution described before fails to fit the shape of the laboratory angular distribution. The width of this angular distribution can fit the experimental width by raising V_a to 4 kcal/mole, but the experiment shows no dip in the center of the distribution and a somewhat slower falloff on the edges of the distribution. The drop in the center of the distribution is mainly due to lack of products with low recoil velocities in the RRKM calculation, not because of insufficient averaging of the RRKM calculation over velocity and angular resolution of the experiment. Actually, the spread in c.m. velocity vector in the $\text{F} + \text{C}_2\text{D}_4$ experiment is only a small fraction of the laboratory angular distribution of $\text{C}_2\text{D}_3\text{F}$. Refinements in estimating the RRKM density of states, such as better estimates of complex frequencies or treatment of the interaction of rotation with vibration as the moments of inertia of the complex change (a small effect because of the light departing atom), may modify slightly the RRKM recoil energy distribution

shown in Fig. 6. After transformation of the c.m. energy distribution to the laboratory angular distribution with experimental averaging, however, such modifications will have no noticeable effects.

The laboratory angular distributions calculated from Eq. (4), on the other hand, fit the experimental curves very closely. The functional form of Eq. (4) is actually a generalization of Eq. (7), with the dimension of the reaction coordinate, n , and the number of vibrational degrees of freedom of the activated complex, s (not including the reaction coordinate), left adjustable. In Eq. (4), however, $(E_{tot} - E')^{s-1}$ is proportional to the "classical" density of states if the zero point energy is not considered to be interchangeable among the oscillators,⁴⁰ whereas $N_r(E_{tot} - E')$ in Eq. (7) is an approximation to the quantum density of states. There is a drastic difference in these systems between the classical and quantum densities since E_{tot} , the energy of the complex above the product zero point level, is less than $\frac{1}{2}$ of the zero point energy of the complex. Thus, the RRKM recoil-energy distributions for C_2H_3F or C_2D_3F can be approximated by Eq. (4) if $n=1$ and s is only $\sim 5-6$.⁴¹ In order to make the number of degrees of freedom variable, the classical density of states was used in Eq. (4) since the quantum density of states depends upon exactly which degrees of freedom are to be excluded, which is not known *a priori*.

The best fit values of s obtained should not be compared with the total number of degrees of freedom (14), but rather with the reduced number ($\sim 5-6$) which is small because of the low energy and the presence of several large vibrational frequencies. The fact that the optimum values of s are only 1.9 for $C_2H_4F^+$ and 1.7 for $C_2D_4F^+$ suggests that there may be incomplete mixing of energy among all degrees of freedom of the complex. Incomplete mixing of energy in this context does not necessarily mean that part of the molecule is isolated for the entire lifetime of the complex, but rather could involve reduced interaction of the reaction coordinate with other degrees of freedom right before the transition state is reached. Such a situation could go unnoticed in experiments concerned only with complex lifetime, but should be detectable by recoil-energy measurements. One should be cautious, however, about taking the values of s too literally. If V_a were known to have some nonzero value, then higher values of s would be necessary in order to fit the experimental curves using Eq. (4) for $P(E')$.

One reason for arguing that energy equilibration in the complex is essentially complete, even though the RRKM curves do not fit the experiments well, is that $C_2H_4F^+$ and $C_2D_4F^+$ decompositions are fit using Eq. (4) with nearly the same value of s , despite the vastly different frequencies involved. If energy equilibration were incomplete, then the relative ease of relaxation of high and low frequencies should be reflected in different product recoil energies.

TABLE I. Best fit recoil-energy parameter s and average recoil-energy $\langle E' \rangle$ for a series of F for H substitution reactions.

Olefin	Optimum s	$\langle E' \rangle$ (kcal/mole)
C_2H_4	1.9	6.7
C_2H_3F	2.3	6.0
C_3H_4 (allene)	3.9	4.4
C_3H_6 (propylene)	2.5	5.8
C_4H_6 (butadiene)	4.3	4.2
C_4H_8 (<i>cis</i> -2-butene)	3.7	4.6
C_4H_8 (<i>trans</i> -2-butene)	3.9	4.4
C_5H_8 (1,3-pentadiene)	4.2	4.3

Another reason for believing that energy is equilibrated to a large extent in the complex is the observation that the fraction of product energy in vibration does increase as the molecular size is increased. This trend was discovered by performing experiments similar to $F + \text{ethylene}$ but with ethylene replaced by fluoroethylene, propylene, allene, 2-butene, butadiene, and 1,3-pentadiene. The resulting angular distributions for H emission products were fit using Eq. (4) for the recoil-energy distribution, with $n=3$, $V_a=0.0$, and the same exoergicity as in Reaction (1). Table I shows the optimum values for s along with the average recoil energy for each system. In general, s is seen to increase with more degrees of freedom, but there are some inconsistencies. Allene, for example, is best fit by $s=3.9$, but propylene by $s=2.5$. Similarly, butadiene has a larger value of s than 2-butene. These deviations from the qualitative predictions of a statistical model, however, may arise from the simplifying assumption that the exoergicity and V_a are the same for all of the substitution reactions in Table I.

Comparison of the energy distribution of product molecules calculated by using RRKM theory and the energy distribution obtained by fitting our experimental results also raises some questions on the applicability of the assumption made in RRKM theory of a unique transition state with a one-dimensional reaction coordinate. When the further assumption is made that the reaction coordinate remains separable after the transition state, then $P(E')$ in Eq. (7) must rise from zero to its maximum value at $E' = V_a$, which is not compatible with the experimental results. Although a small contribution of orbital angular momentum to the effective potential in the reaction coordinate at the transition state, and mixing of recoil energy with rotation of C_2H_3F because of a finite exit impact parameter of separation may smooth out somewhat an initially sharp recoil energy distribution, these effects will not be large enough, especially in these reactions, to improve significantly the fit to experiment using the RRKM recoil-energy distribution.⁴² If the mixing of energy among different modes of motion is complete in

(C₂H₄F)[†], our experimental results would strongly suggest the inadequacy of the transition state assumption, even in the decomposition of a long-lived complex. The complicated process of decomposition of a polyatomic molecule evidently cannot be described by a unique transition state with a single value of V_a . The actual reaction path might correspond to a wide range of transition states with various values of V_a and may consist of more than one degree of freedom at any of these transition states.

B. Abstraction Reaction

In the abstraction reaction (3), the formation of a substantial amount of C₂D₃ with very little internal excitation implies that the C–D bond breakage is quite adiabatic with respect to C₂D₃ excitation. That is, the C₂D₃ has time to adjust its coordinates smoothly to final values while the D is being removed. This adiabaticity is to be expected if the C–D bond breakage occurs when F is still somewhat further away from D than the C–D distance, which is supported by the high vibrational excitation of the DF formed. The sharp forward and backward peaks in the angular distribution of DF indicate that the final orbital angular momentum is a large fraction of the total angular momentum, which is to be expected when a light atom is transferred between two heavy molecules. The presence of both forward and backward peaks suggests that a wide range of impact parameters contribute to the abstraction reaction, or else that the strong attractive forces in the separation channel keep the polar DF close to the free radical C₂D₃ long enough for rotation.

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